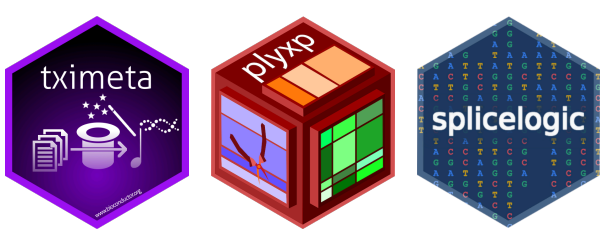
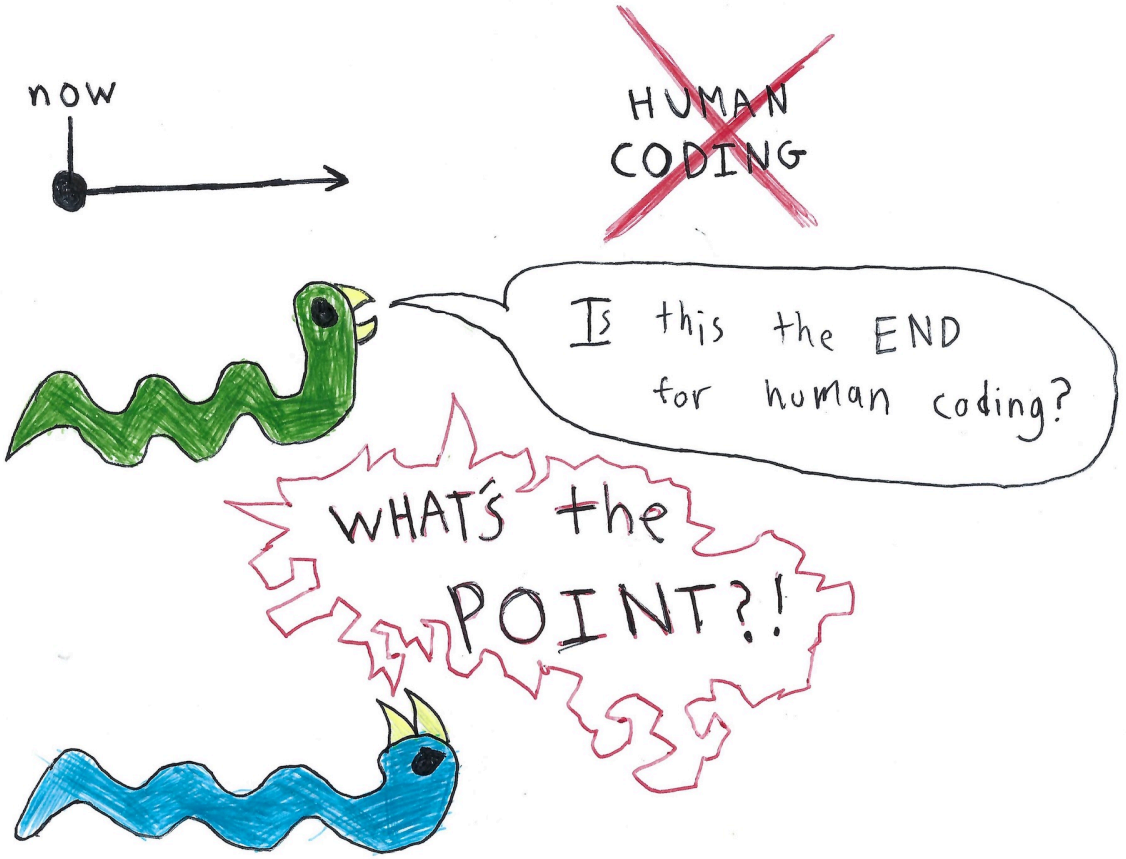
Frameworks for multi-sample long-read RNA-seq analysis

Michael Love, Justin Landis, Beatriz Campillo, Rachel Sharp

UNC-Chapel Hill, Department of Genetics
Biological Data Science 2026





- An LLM writes good R/python code for RNA-seq in seconds
- Why build frameworks and new syntax for doing functional genomics?

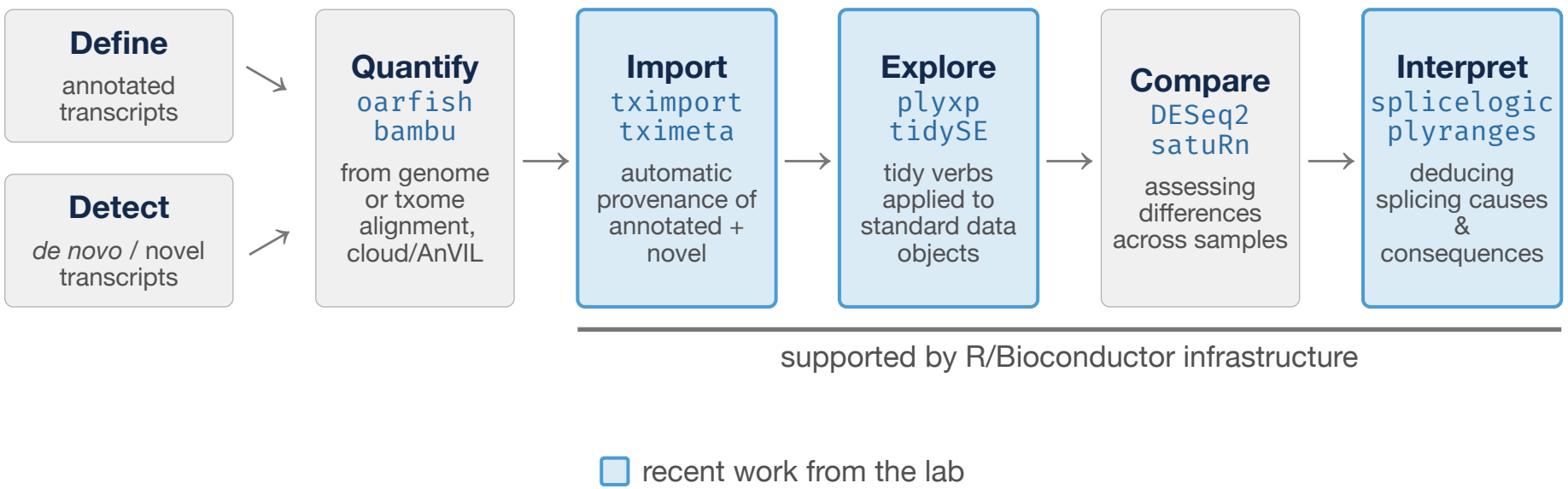
Why new syntax (for humans or LLMs)?

1. Humans read and review code (for now)
   Concise and easy to comprehend
2. Grammar enables precision while maintaining readability, e.g.
   “find peaks near DE genes” vs.

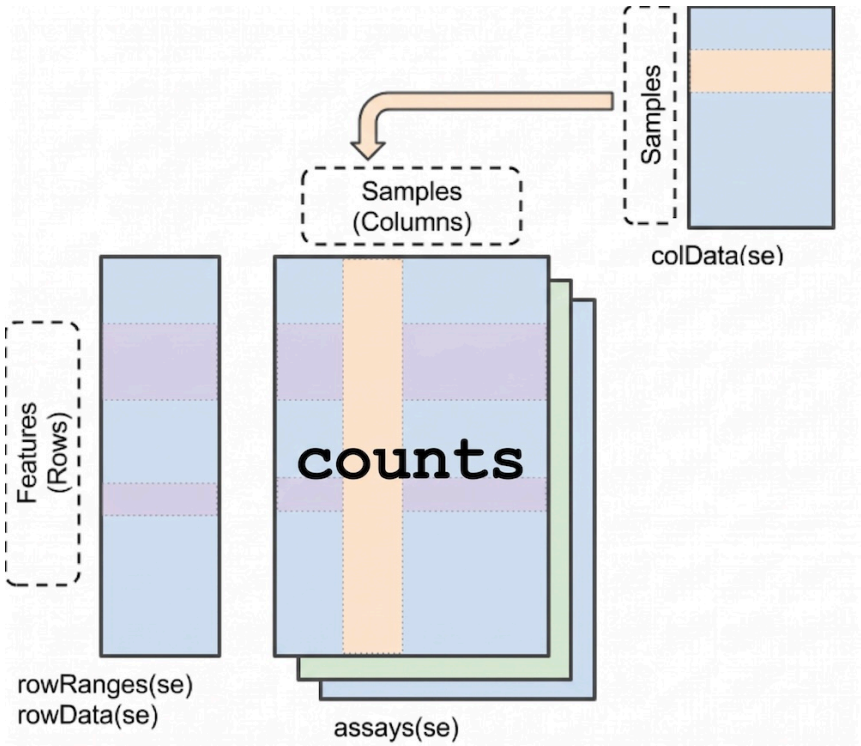
```
genes ▷ filter(de) ▷ promoters( ... ) ▷ join_overlap_inner(peaks, ... )
```

3. Composable verbs map meaningfully to many analyses
   With standard data classes, catches silent bookkeeping & genomics errors
4. Tested, maintained verbs safer than freshly generated code
   Same verbs work consistently across backends

Multi-sample long-read RNA-seq workflow



Object-oriented genomics



- *SummarizedExperiment* links matrix to feature and sample metadata
- Analagous to python's *anndata* object

Transcript metadata complications



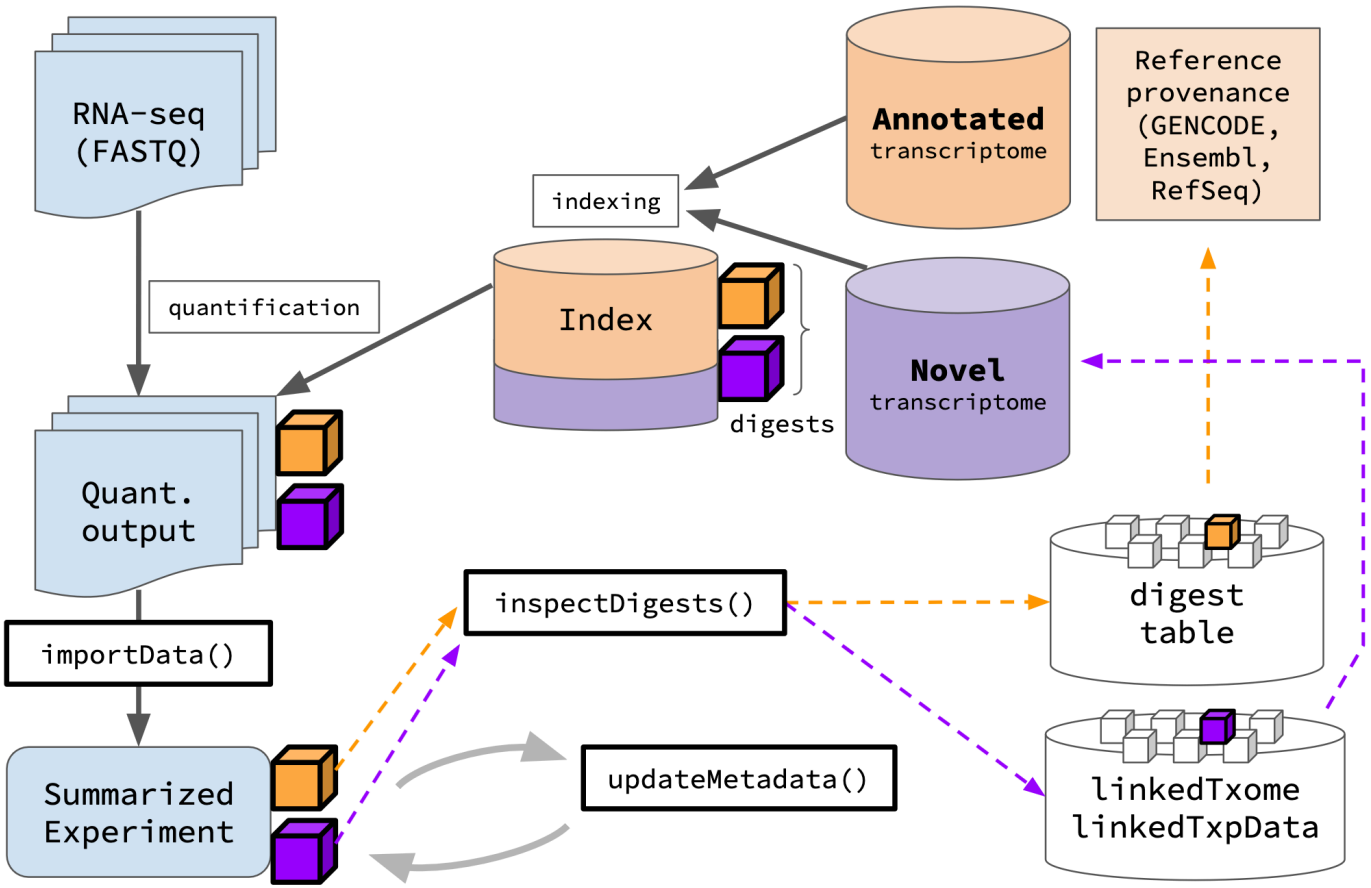

```
chr10 de-novo-tool transcript 3779047 3785193 . - .
  gene_id "ENSG00000067082.14"; transcript_id "ENST00000497571.5";
chr10 de-novo-tool exon 3779047 3779590 . - 0 ...
chr10 de-novo-tool exon 3780106 3780229 . - 0 ...
chr10 de-novo-tool exon 3781641 3782214 . - 0 ...
chr10 de-novo-tool transcript 5094414 5107680 . + .
  gene_id "ENSG00000196139.13"; transcript_id "ENST00000380554.4";
chr10 de-novo-tool exon 5094414 5094528 . + 0 ...
chr10 de-novo-tool exon 5096410 5096577 . + 0 ...
chr10 de-novo-tool exon 5097434 5097550 . + 0 ...

chr10 de-novo-tool transcript 89293545 89535556 . - .
  gene_id "chr10:89293000";
  transcript_id "8327445e-4522-47f2-a2dd-af1b031c1917_chr10:89293000";
chr10 de-novo-tool exon 89293545 89293668 . - 0 ...
chr10 de-novo-tool exon 89412791 89412922 . - 0 ...
chr10 de-novo-tool exon 89436060 89436319 . - 0 ...
chrY de-novo-tool transcript 12904831 12920478 . + .
  gene_id "ENSG00000067048.16";
  transcript_id "141285fb-d028-4740-a9d9-f04a28fbaef1";
chrY de-novo-tool exon 12904831 12904981 . + 4879 ...
chrY de-novo-tool exon 12907537 12907594 . + 4879 ...
chrY de-novo-tool exon 12909360 12909407 . + 4879 ...
```

Annotated

Novel

tximeta: mixed reference workflow (2025)



`tximeta`: inspect, link, and update metadata




```
se ← importData(coldata, type = "oarfish")
inspectDigests(se)

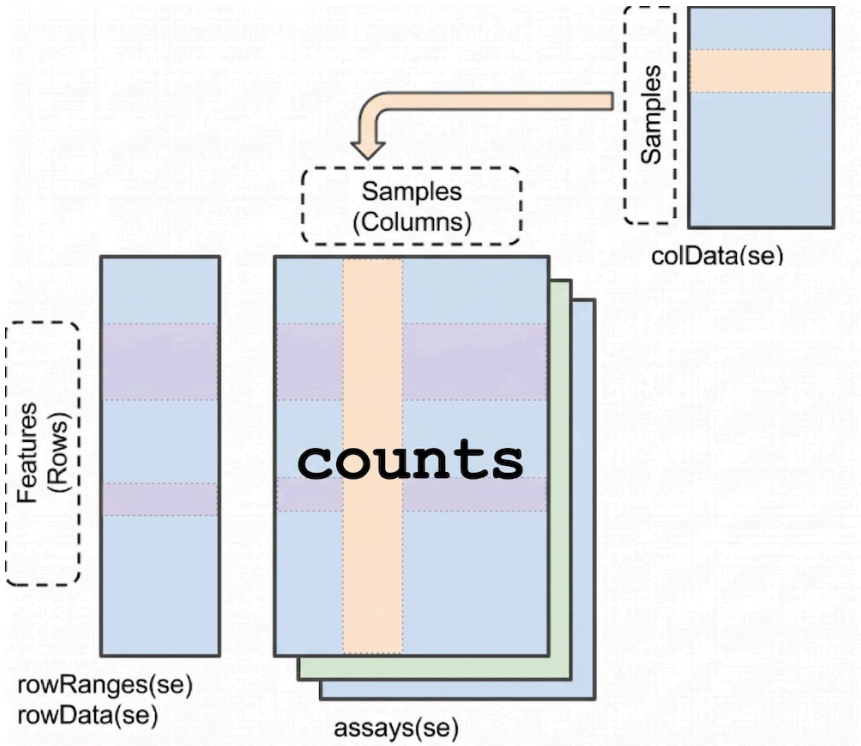
#> index     source  organism release genome  ...
#> annotated GENCODE Homo sa… 48      GRCh38  ...
#> novel     <NA>    <NA>     <NA>    <NA>    ...

# <add or link metadata in local cache>
se ← updateMetadata(se, ranges = TRUE)
```



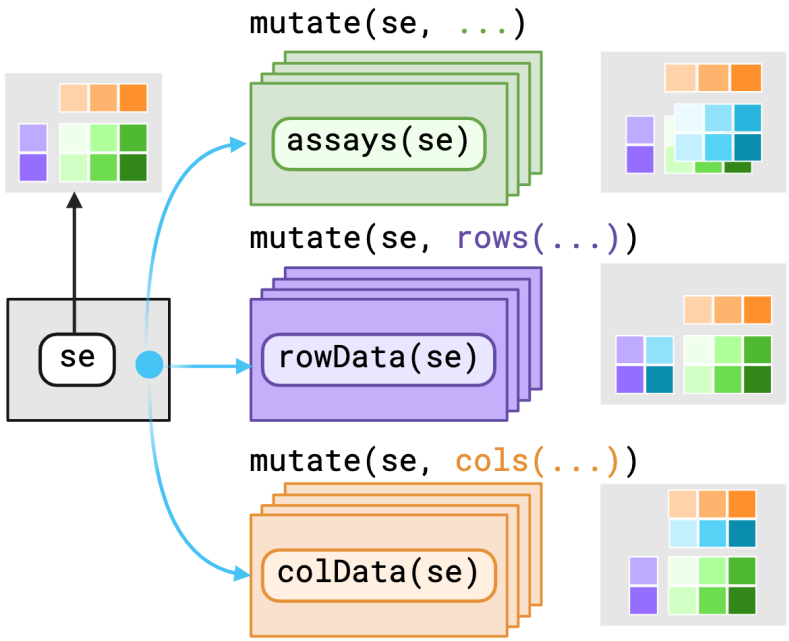
Works for `salmon` and other quantifiers too

You have your object, now what?



- Even simple exploration requires learning domain-specific methods
- (AI-generated) code can quickly become obtuse

A grammar for annotated matrices

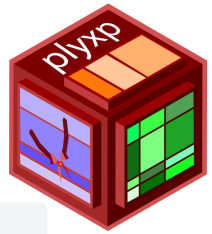


- `plyxp` brings *dplyr* verbs to annotated matrices, efficiently
- Expressions are scoped to contexts: `assays`, `rows()`, or `cols()`
- Pronouns `.assays`, `.rows`, `.cols` reach across contexts
- We are looking to support *AnnData*



Justin Landis

plyxp: mutate, filter, aggregate

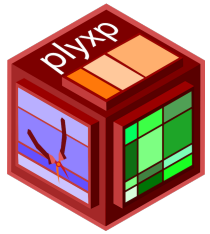



```
xp ← new_plxyp(se) ▷
  mutate(
    scaled_cts = counts / .cols$size_factor,
    rows(ave = map_dbl(.assays$scaled_cts, mean))
  ) ▷
  filter(rows(ave > 10), cols(batch ≠ "b3"))

# also: rowMeans(.assays_asis$scaled_cts)

genes ← xp ▷
  group_by(rows(gene_id)) ▷
  summarize(counts = colSums(counts))
```

FP control for aggregation: count splitting

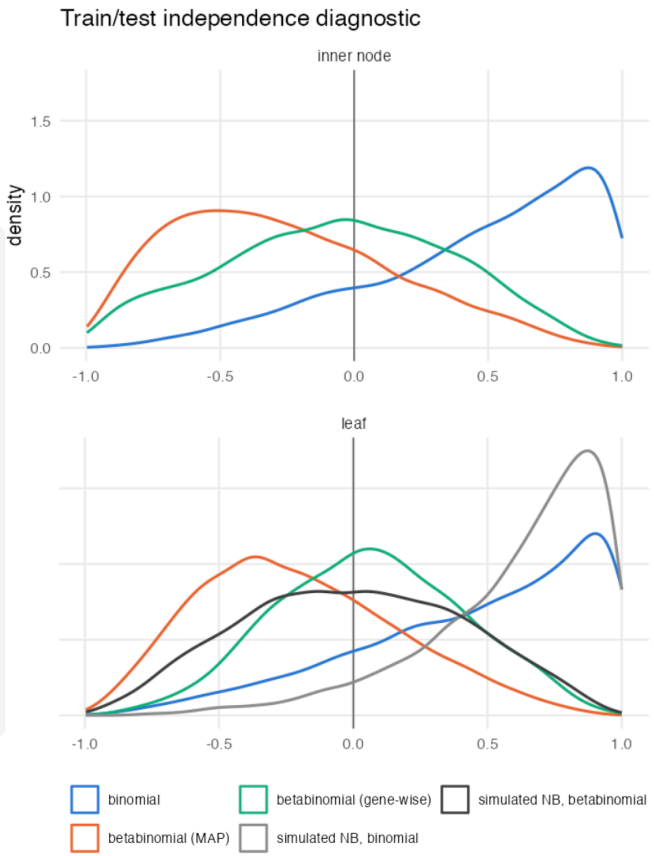


Aggregate counts from similar transcripts:

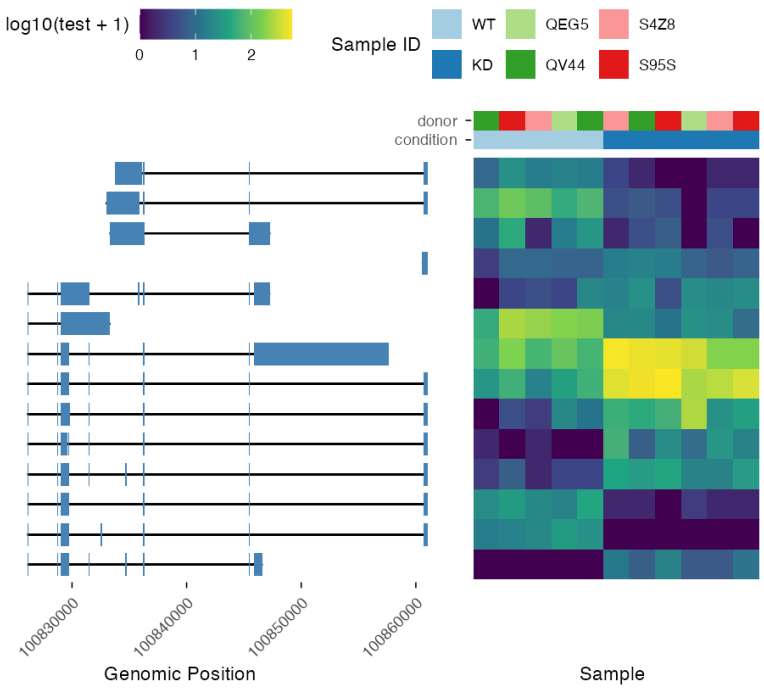
```
split ← xp ▷
  mutate(train = rbinom(n(), counts, 0.5),
         test = counts - train)

tree ← split ▷
  group_by(rows(gene_id)) ▷
  mutate(rows(nodes = exons ▷ cluster_data()))

inner_nodes ← tree ▷
  group_by_nodes(rows(inner_nodes(nodes))) ▷
  summarize(across(everything(), colSums))
```



Transcript usage across samples



- Splicing changes rarely come one event at a time
- How to turn transcript sets into events?
- … into biological interpretation?



Beatriz Campillo

splicelogic: events as composable ranges

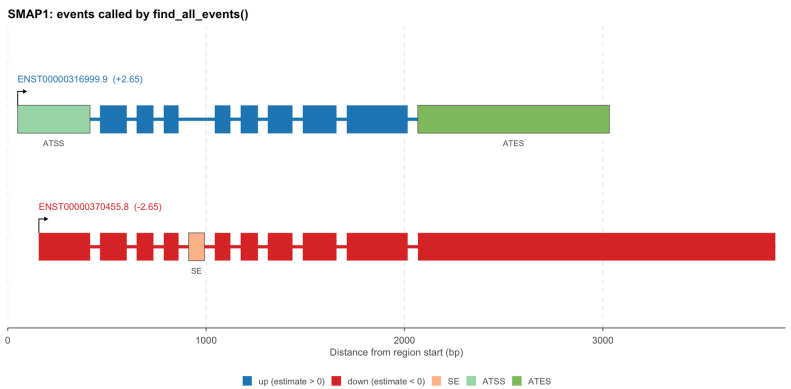


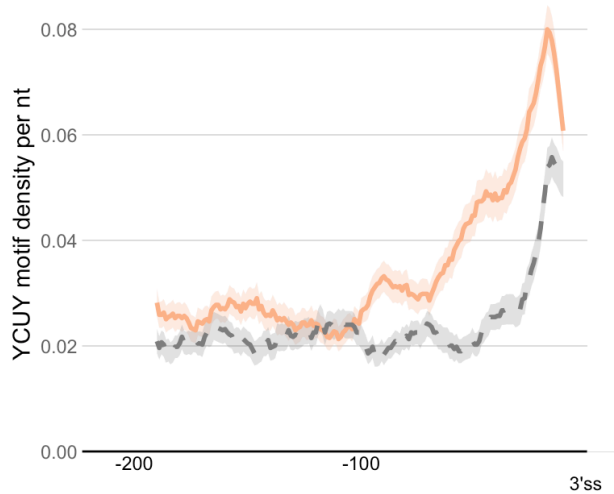

```
# <ranges> to <ranges>: "endomorphic"
events ← exons ▷ find_all_events()

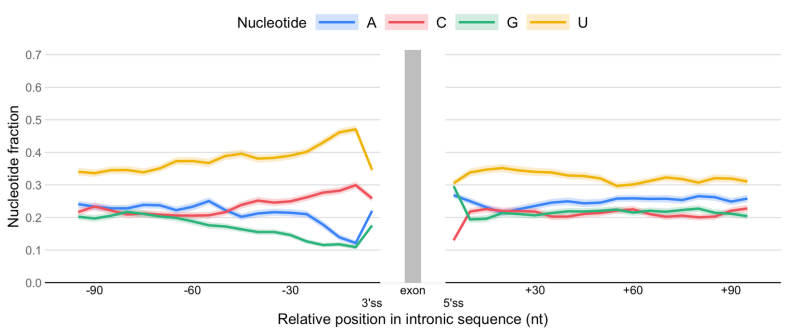
skipped ← exons ▷
  find_skipped_exons() ▷
  flank_upstream(100) %>%
  mutate(intron_seq = get_seq(., "hg38"))

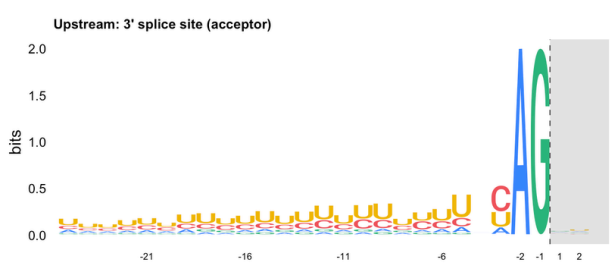
# results attached to ranges as columns
skipped$intron_seq
```

splicelogic: GTEx PTBP1 knockdown









Reanalysis of: Glinos *et al.*, (GTEX v9) *Nature* 608, 353–359 (2022), doi:10.1038/s41586-022-05035-y

analyzed on

Modular tools and functions



TAPIR: transcript analysis pipelines in R

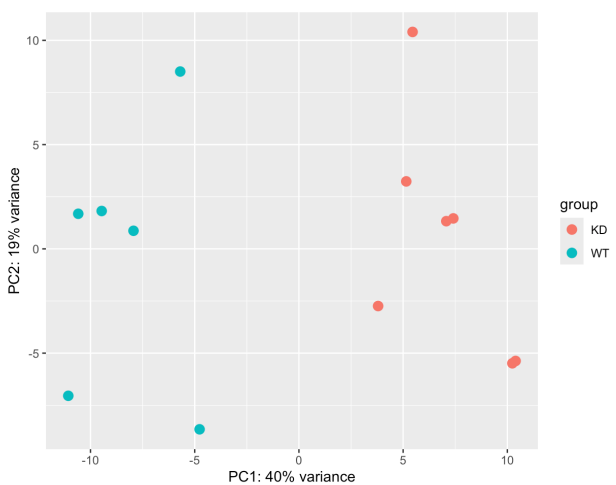
TAPIR: Transcript analysis pipelines in R

Book contents
1 Introduction
2 Exploratory data analysis
3 Statistical testing
4 Post DTU Analysis
5 Event Annotation with plyxp + splicelogic
References

PCA plot:

```
vsd <- vst(dds, blind=FALSE)
plotPCA(vsd)
```

```
using ntop=500 top features by variance
```

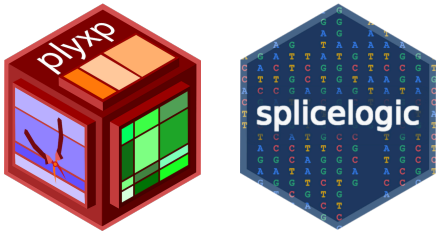


- Collaboration with Stephanie Hicks (JHU), Sowmya Parthiban (JHU)
- Chapters are being re-organized now by *Dr. Rachel Sharp*
- `txomics/tapir`



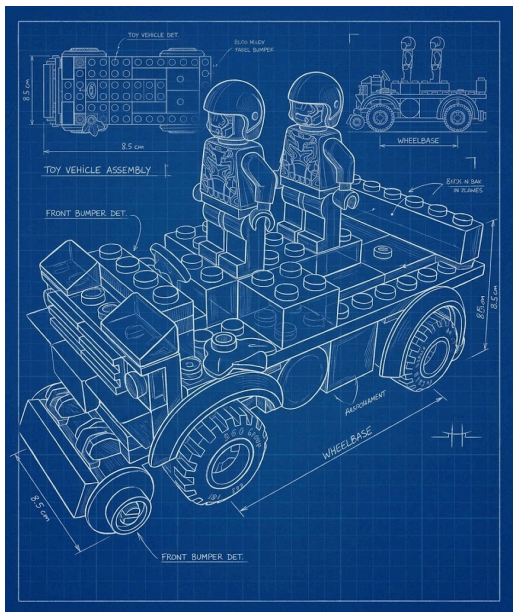
Rachel Sharp

Compute and propagate custom stats



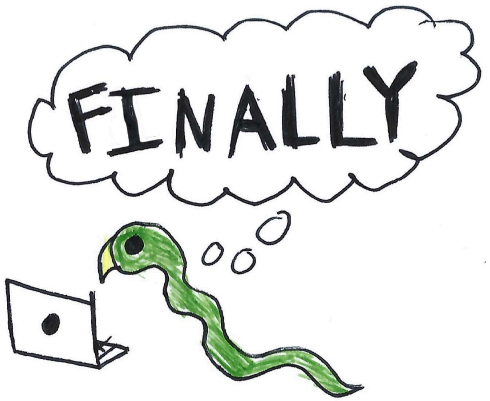

```
usage_per_group ← xp ▷
  group_by(cols(condition)) ▷
  summarize(
    mean = rowMeans(usage),
    sd = rowSds(usage))

# to propagate custom stats to events
exons ▷
  preprocess( ... ,
    additional_columns = ... )
```

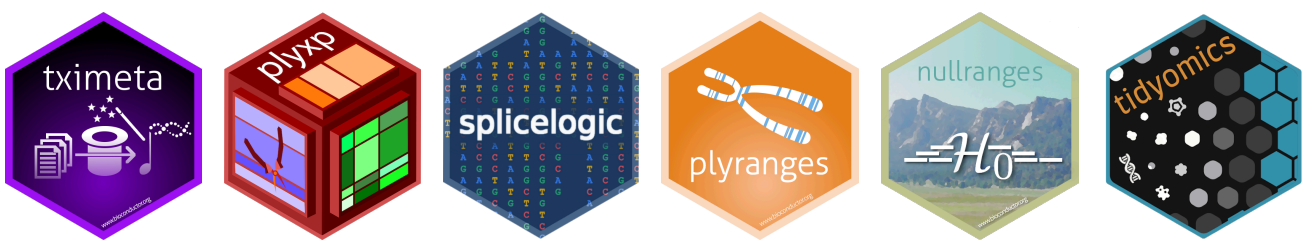


View at txomics.github.io/tapir/annotate-plyxp-splicelogic

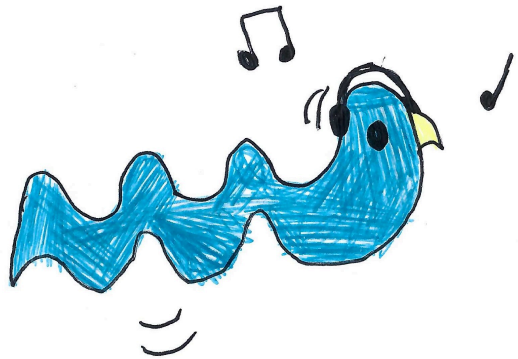
Back to the question: why new syntax?



- `tximeta`: provenance travels with the data
- `plyxp`: *dplyr* verbs on standard objects
- `splicelogic`: ranges in, ranges out: composes with `plyranges`
- Tested verbs and classes handle the bookkeeping



Acknowledgments



Love lab, UNC-Chapel Hill

- Justin Landis — `plyxp`
- Beatriz Campillo — `splicelogic`
- Rachel Sharp — `tapir`
- Isaiah Love — “florms” 🦪

Collaborators

- Zahra Zare Jousheghani — `oarfish`
- Rob Patro and Patro lab members
- Charlotte Soneson — `tximeta`
- Stefano Mangiola et al. — `tidyomics`

Software links

- github.com/thelovelab
- github.com/tidyomics
- mikelove.github.io ▷ software
- txomics/tapir

Funding

- NIH / NHGRI R01-HG009937
- Wellcome Trust, Essential Open Source Software for Science (EOSS) 313919/Z/24/Z


Beatriz


Justin


Rachel


Zahra